



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to PALANIMALAI TRADERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

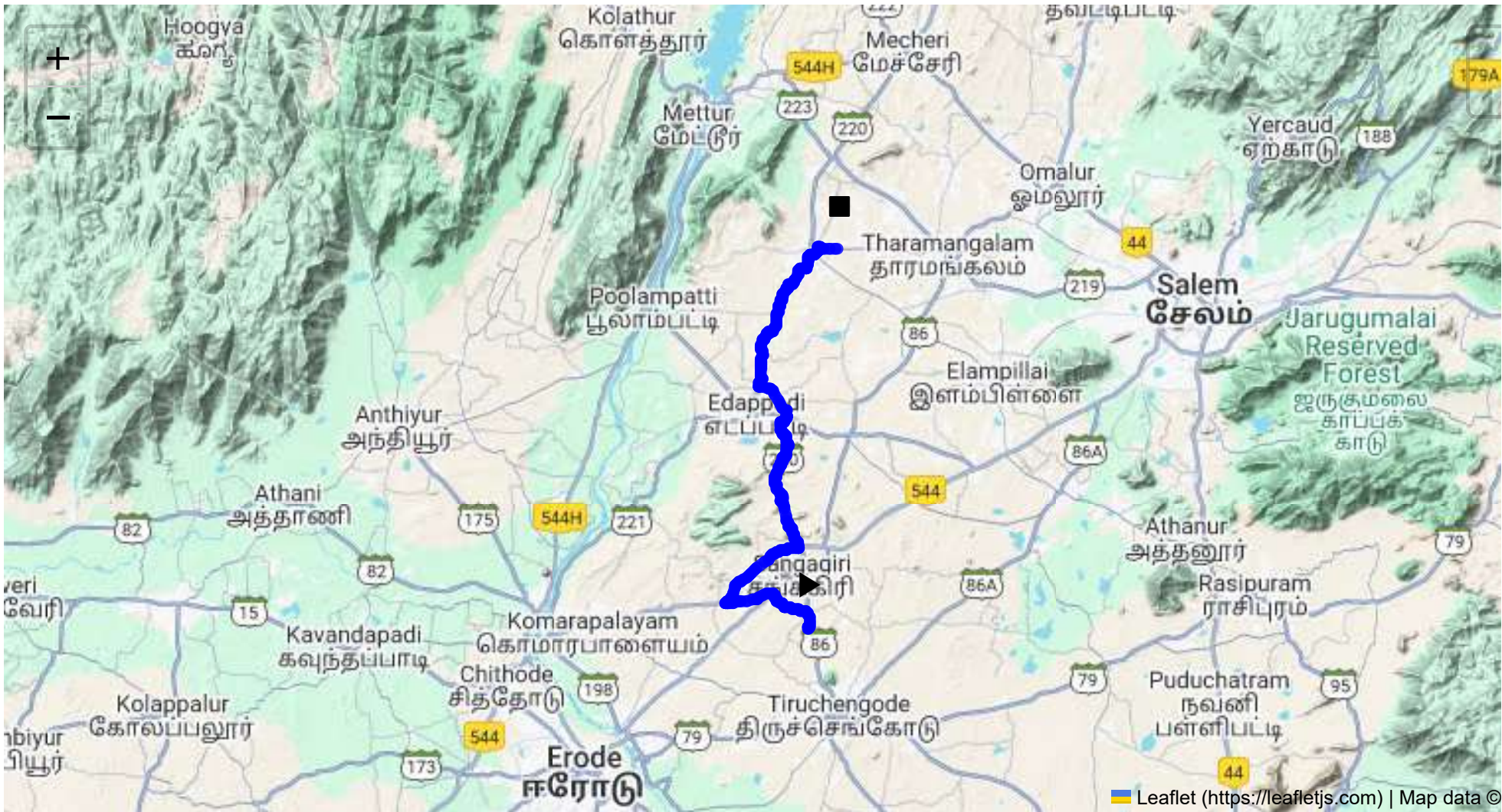
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 44.73 km
Estimated Duration: 1.1 hours
Adjusted Duration (Heavy Vehicle): 1.4 hours
Start: (11.4381, 77.8734)
End: (11.69724708, 77.8950329)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map:

- The route starts from Sangagiri, Tamil Nadu and ends at Surappalli, Tamil Nadu, with key waypoints at Sankari RS - Erode Privu Road, Bhavani Main Road through Sankari West and Padaiveedu.

- The total distance is approximately 44.73 kilometers, which is expected to take around 1.09 hours for heavy vehicles carrying hazardous materials.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- Tamil Nadu generally experiences a tropical climate. During monsoon seasons (June to September and October to November), heavy rainfall can lead to water-logged streets and reduced visibility.
- High temperatures (up to 40°C) in summer (March to May) may impact vehicle performance, particularly those with sensitive materials.

3. Analysis of Traffic Patterns, Highlighting Peak Hours and Congestion-Prone Areas:

- Morning (8 AM - 10 AM) and evening (5 PM - 8 PM) are typically peak traffic hours.
- Notable congestion areas include the junctions at Sankari RS and the market zones in Sankari West and Padaiveedu.

4. Assessment of Road Quality and Infrastructure:

- The main roads (Bhavani Main Road and Sankari RS - Erode Privu Road) are generally in good condition. However, some minor roads may have potholes and uneven surfaces.
- Regular maintenance and roadwork could cause temporary detours and/or delays.

5. Suggestions for Alternative Routes for Emergencies:

- Alternate routes include utilizing smaller connecting roads to avoid major traffic blocks; however, these may increase transit time.
- In emergencies, connecting back to NH-44 provides a larger, more accessible thoroughfare.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Movement of hazardous materials usually requires special permits and compliance with Tamil Nadu safety regulations, including specific time frames for transportation to minimize risk.
- Restrictions are heightened around populated and environmentally sensitive areas.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:

- Past incidents in this region are rare but typically involve accidents due to human error or weather conditions such as slippery roads.
- Proper training and adherence to safety protocols are critical.

8. Environmental Considerations and Sensitive Areas:

- Parts of the route may pass near agricultural lands and residential areas. Attention to spillage risk and minimal noise pollution is vital.
- Ecologically sensitive areas should be navigated with caution to prevent environmental damage.

9. Analysis of Communication Coverage, Noting Potential Dead Zones:

- While most areas along the main route have good telecommunication coverage, certain rural segments might experience network issues.
- Equipped communication systems like satellite phones can aid in overcoming dead zones.

10. Estimated Emergency Response Times for Different Route Segments:

- Emergency response in urban areas like Sangagiri and Surappalli can be prompt, typically within 10-20 minutes.
- More rural stretches, particularly around Padaiveedu, may experience delays up to 30-45 minutes.

11. Overall Summary of Risk Assessment:

- This route, while generally straightforward, presents certain risks related to weather, traffic congestion, and the handling of hazardous materials.
- Adequate preparation, awareness of peak traffic hours, alternative routes, and communication plans are crucial to ensure safety.
- Continued monitoring of road conditions, weather patterns, and local traffic regulations will contribute significantly to reducing transport risks along this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60348, 77.83967	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.44900, 77.87406	30 KM/Hr
6	Turn	Medium	11.45357, 77.85789	30 KM/Hr
7	Turn	Medium	11.45352, 77.85698	30 KM/Hr
8	Turn	Medium	11.45847, 77.85140	30 KM/Hr
9	Turn	Medium	11.46048, 77.85036	30 KM/Hr
10	Turn	Medium	11.46303, 77.84945	30 KM/Hr
11	Turn	Medium	11.46314, 77.84928	30 KM/Hr
12	Turn	High	11.45542, 77.81725	15 KM/Hr
13	Turn	High	11.45634, 77.81673	15 KM/Hr
14	Blind Spot	Blind Spot	11.49390, 77.86748	10 KM/Hr
15	Turn	Medium	11.49404, 77.86724	30 KM/Hr
16	Turn	Medium	11.50866, 77.86053	30 KM/Hr
17	Turn	Medium	11.55039, 77.85511	30 KM/Hr
18	Turn	Medium	11.55460, 77.85624	30 KM/Hr
19	Turn	High	11.57893, 77.85430	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
20	Turn	Medium	11.57903, 77.85441	30 KM/Hr
21	Turn	Medium	11.58022, 77.85744	30 KM/Hr
22	Turn	Medium	11.60727, 77.84032	30 KM/Hr
23	Turn	Medium	11.60972, 77.84164	30 KM/Hr
24	Turn	Medium	11.64064, 77.84611	30 KM/Hr
25	Turn	Medium	11.64535, 77.85122	30 KM/Hr
26	Turn	Medium	11.66467, 77.85599	30 KM/Hr
27	Turn	Medium	11.67301, 77.86024	30 KM/Hr
28	Turn	Medium	11.67326, 77.86058	30 KM/Hr
29	Turn	Medium	11.67373, 77.86230	30 KM/Hr
30	Turn	Medium	11.68171, 77.86676	30 KM/Hr
31	Turn	High	11.68490, 77.86893	15 KM/Hr
32	Turn	High	11.68472, 77.87281	15 KM/Hr
33	Turn	Medium	11.69280, 77.87439	30 KM/Hr
34	Turn	Medium	11.69458, 77.87589	30 KM/Hr
35	Blind Spot	Blind Spot	11.69360, 77.87831	10 KM/Hr
36	Turn	Medium	11.69523, 77.87892	30 KM/Hr
37	Turn	High	11.69918, 77.88105	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Parusuraman Hospital	11.697164, 77.877792	30 km/h	Medium
2	clinic	Rajkumar Clinic	11.695372, 77.879705	30 km/h	Medium
3	hospital	Nakul Hospital	11.6954119, 77.8796544	30 km/h	Medium
4	hospital	Rajeshwaran Hospital	11.696836, 77.879517	30 km/h	Medium
5	hospital	Government Hospital, Jalakandapuram	11.6998786, 77.8792017	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58300, 77.85813



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60348, 77.83967



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46048, 77.85036



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46314, 77.84928



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45634, 77.81673



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49390, 77.86748



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49404, 77.86724



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50866, 77.86053



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55039, 77.85511



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55460, 77.85624



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57893, 77.85430



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57903, 77.85441



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58022, 77.85744



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60727, 77.84032



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60972, 77.84164



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64064, 77.84611



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64535, 77.85122



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66467, 77.85599



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67301, 77.86024



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67326, 77.86058



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67373, 77.86230



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68171, 77.86676



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.68490, 77.86893



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.68472, 77.87281



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69280, 77.87439



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69458, 77.87589



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.69360, 77.87831



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69523, 77.87892



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69918, 77.88105